



Introduction to R

As you grow as a data analyst, you may need to learn additional programming languages. This program teaches Python, which is the most widely used language for data analysis. Python is a great starting point for foundational analysis, and its structure and syntax make it an accessible language for beginners.

Another powerful programming language for data analysis is R. R is typically used by professionals with a statistical or research-oriented approach to problem-solving, such as data scientists, statisticians, and engineers. This reading provides a general introduction to R, its uses, and its benefits for data analysis. Knowing about R is helpful since you may encounter the language in a future professional role.

Note that many of the coding concepts you'll learn with Python in this course are transferable to other programming languages, making it easier to pick up tools like R as you progress in your career.

What is R?

Historical context

R is an open-source programming language and software environment specifically designed for statistical computing and graphics. Created in 1993 by professors Ross Ihaka and Robert Gentleman at the University of Auckland, New Zealand, R has since become a standard tool in data science, academic research, and various industries.

Unlike general-purpose languages such as Python or Java, R is a domain-specific language whose features and syntax are tailored for statistical and data-related tasks. Most R users work within an Integrated Development Environment (IDE) called RStudio, which provides a user-friendly interface for writing code, viewing plots, and managing data. You can think of R as a comprehensive toolbox for working with data, built to handle everything from simple calculations to complex statistical modeling.

Uses and applications

R's primary use is in the fields of data analysis and data science.

- Statistical analysis: R is a standard tool for statistical analysis in academia and industry. Researchers use it to conduct complex statistical tests, build predictive models, and validate hypotheses.
- Machine learning: R has a rich ecosystem of packages for machine learning tasks, from simple linear regression to advanced deep learning. Packages such as caret and tidymodels offer a unified interface for building, training, and evaluating models.
- Data wrangling: A significant portion of any data analysis project involves cleaning and preparing data. R has powerful tools for this, most notably the tidyverse suite of packages. These packages, including dplyr for data manipulation and tidyr for data tidying, facilitate the process of transforming raw data into a usable format.
- Data visualization: R excels at creating high-quality, professional plots and charts. Packages like ggplot2 can help you generate informative visualizations to explore data and communicate key findings.
- Reporting: R can be integrated with tools like R Markdown to create dynamic reports, presentations, and dashboards that combine code, output, and prose. This ensures that reports are always up-to-date with the latest data and analysis.

Benefits

R offers several key benefits that make it a useful tool for data analysts and data scientists.

- Open-source: R is a free, open-source project. This means that anyone can use R without licensing fees. Moreover, a global community of users and developers continuously contributes to R's improvement and expands its capabilities.
- Vast ecosystem of packages: R has an extensive library of packages available on the Comprehensive R Archive Network (CRAN). With over 20,000 packages, R can handle almost any data analysis task, from specialized genomics analysis to financial modeling.
- Dedicated to data: R was built by and for statisticians, so its design centers on data. R supports working with data structures and syntax that streamline the entire data analysis process.
- Strong community support: The R community is large, active, and highly collaborative. This makes it easy for R users to find help, tutorials, and solutions to problems by accessing the R community's forums, blogs, and other online resources.

Key takeaways

R is a robust and flexible programming language for data analysis. R's statistical capabilities, visualization options, and extensive package ecosystem make it a useful tool for data professionals and researchers worldwide.

Additional resources

Check out the following online resources to further explore R:

- [R for data science](#) is a free online book by Hadley Wickham and Garrett Grolemund that includes a comprehensive introduction to R and the tidyverse, a collection of R packages designed for data science. It emphasizes practical data manipulation, visualization, and modeling.
- [RStudio Education](#) provides tutorials, webinars, and learning pathways for beginners to get started with R and RStudio. Many R users work with an integrated development environment (IDE) like RStudio, which provides a user-friendly interface with a code editor, console, and tools for plotting and managing data.
- [CRAN \(Comprehensive R Archive Network\)](#) is the central repository for R packages and offers access to extensive documentation and resources for thousands of R packages.